

BOOTSTRAP

Interview & Reference Cheat Sheet

Grid system, components, utilities & common interview Q&A

1. What is Bootstrap?

- Bootstrap is a free, open-source CSS framework for building responsive, mobile-first websites quickly.
- It provides pre-built, reusable components (buttons, cards, navbars, modals, etc.) and a 12-column responsive grid system.
- It enforces visual consistency and uniformity across a project without writing CSS from scratch.
- Built on HTML, CSS, and (for interactive components) JavaScript.

1.1 Mobile-First Approach

Bootstrap's CSS is written mobile-first: base styles target the smallest screens, and media queries (min-width) progressively add styles as the viewport grows. This means you design for mobile first, then layer on rules for larger breakpoints — not the other way around.

1.2 Ways to Use Bootstrap

Method	Description
CDN	Link Bootstrap's hosted CSS/JS files directly in <code><head>/before </body></code> — fastest way to start, no install needed.
Download	Download the compiled CSS/JS files and host them yourself.
Package manager	Install via npm (npm install bootstrap) and import into a build pipeline (Webpack/Vite) for customization with Sass.

Using the CDN

```
<!-- In <head> -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"
rel="stylesheet">

<!-- Before closing </body> (needed for dropdowns, modals, carousel, etc.) -->
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/bootstrap.bundle.min.js"><
/script>
```

2. The Grid System

Bootstrap's layout is built on a 12-column grid using three levels of elements:

- Container — wraps and centers content, sets a max-width (or full width for -fluid)
- Row — a horizontal wrapper for columns (uses negative margins to offset column padding)
- Column(s) — the actual content areas; up to 12 columns can sit in one row

```
<div class="container">
  <div class="row">
    <div class="col-6">Half width</div>
    <div class="col-6">Half width</div>
  </div>
</div>
```

2.1 Container Types

Class	Behavior
container	Fixed, responsive max-width at each breakpoint (centered, with side padding)
container-fluid	Always 100% of viewport width at every breakpoint
container-{breakpoint}	e.g. container-md — 100% wide until that breakpoint, then fixed-width like container

2.2 Breakpoints

These are the standard Bootstrap 5 breakpoints (min-width triggers):

Breakpoint	Class infix	Min width	Typical device
Extra small	(none)	<576px	Phones (portrait)
Small	sm	≥576px	Phones (landscape)
Medium	md	≥768px	Tablets
Large	lg	≥992px	Laptops
Extra large	xl	≥1200px	Desktops
Extra extra large	xxl	≥1400px	Large desktops

Note on your original notes: sm is ≥576px, md is ≥768px, lg is ≥992px, and xl is ≥1200px — these thresholds are correct in your notes; I've just reorganized and confirmed them here.

2.3 Column Classes

```
<div class="row">
  <div class="col-4">col-4</div>      <!-- always 4/12 wide -->
  <div class="col-md-4">col-md-4</div> <!-- 4/12 wide from md up, full width below -->
  <div class="col">col</div>          <!-- auto-sized, equal width -->
```

```
</div>
```

Column classes can be combined per breakpoint for responsive layouts, e.g. `class="col-12 col-sm-6 col-lg-4"` — full width on mobile, half on small screens, one-third on large screens.

3. Core Components

3.1 Buttons

<pre><button type="button" class="btn btn-primary">Primary</button> <button type="button" class="btn btn-secondary">Secondary</button> <button type="button" class="btn btn-success">Success</button> <button type="button" class="btn btn-danger">Danger</button> <button type="button" class="btn btn-warning">Warning</button> <button type="button" class="btn btn-info">Info</button> <button type="button" class="btn btn-link">Link</button> <button type="button" class="btn btn-outline-primary">Outline</button></pre>	
Class	Color meaning
btn-primary	Main brand blue
btn-secondary	Neutral grey
btn-success	Green — success/confirmation
btn-danger	Red — destructive/error actions
btn-warning	Orange/yellow — caution
btn-info	Light blue — informational
btn-link	Styled to look like a plain link
btn-outline-*	Outlined version of any color variant

3.2 Cards

<pre><div class="card" style="width: 18rem;"> <div class="card-header">Header</div> <div class="card-body"> <h5 class="card-title">Card title</h5> <p class="card-text">Some quick example text.</p> Card link </div> <div class="card-footer text-muted">Footer</div> </div></pre>	
Class	Purpose
card	Outer wrapper
card-header / card-footer	Optional top/bottom sections
card-body	Main padded content area
card-title / card-text	Heading and body text inside card-body
card-img-top	Image spanning the top of the card
card-link	A link styled with card-appropriate spacing

3.3 Navbar

```
<nav class="navbar navbar-expand-lg bg-body-tertiary">
  <div class="container-fluid">
    <a class="navbar-brand" href="#">Brand</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarNav">
      <span class="navbar-toggler-icon"></span>
    </button>
    <div class="collapse navbar-collapse" id="navbarNav">
      <ul class="navbar-nav">
        <li class="nav-item"><a class="nav-link active" href="#">Home</a></li>
        <li class="nav-item"><a class="nav-link" href="#">Features</a></li>
      </ul>
    </div>
  </div>
</nav>
```

navbar-expand-lg means the navbar is collapsed into a hamburger menu below the lg breakpoint, and expands into a horizontal menu at lg and above.

3.4 Carousel

A slideshow component for cycling through images or content.

```
<div id="myCarousel" class="carousel slide" data-bs-ride="carousel">
  <div class="carousel-indicators">
    <button data-bs-target="#myCarousel" data-bs-slide-to="0"
class="active"></button>
    <button data-bs-target="#myCarousel" data-bs-slide-to="1"></button>
  </div>
  <div class="carousel-inner">
    <div class="carousel-item active"></div>
    <div class="carousel-item"></div>
  </div>
  <button class="carousel-control-prev" type="button" data-bs-target="#myCarousel"
data-bs-slide="prev">
    <span class="carousel-control-prev-icon"></span>
  </button>
  <button class="carousel-control-next" type="button" data-bs-target="#myCarousel"
data-bs-slide="next">
    <span class="carousel-control-next-icon"></span>
  </button>
</div>
```

Class / attribute	Purpose
carousel slide	Enables the sliding transition (vs. a hard cut)
carousel-inner / carousel-item	Wrapper and individual slides — exactly one item needs class active
carousel-control-prev / -next	Previous/next click controls

carousel-indicators	The small dot/line indicators for jumping to a specific slide
data-bs-ride="carousel"	Auto-starts the slideshow on page load

3.5 Accordion (Collapsible Panels)

This is almost certainly what your note "animation... allows multiple stepper" was referring to — Bootstrap's Accordion component, which lets users expand/collapse stacked content panels, optionally allowing multiple panels open at once.

```
<div class="accordion" id="accordionExample">
  <div class="accordion-item">
    <h2 class="accordion-header">
      <button class="accordion-button" type="button" data-bs-toggle="collapse" data-
bs-target="#collapseOne">
        Item #1
      </button>
    </h2>
    <div id="collapseOne" class="accordion-collapse collapse show" data-bs-
parent="#accordionExample">
      <div class="accordion-body">Panel content goes here.</div>
    </div>
  </div>
</div>
```

Set data-bs-parent on each panel to allow only one panel open at a time (classic accordion behavior). Omit data-bs-parent to allow multiple panels to stay open simultaneously.

3.6 Alerts

```
<div class="alert alert-success" role="alert">
  Success! Your changes were saved.
</div>

<div class="alert alert-danger alert-dismissible fade show" role="alert">
  Error message here.
  <button type="button" class="btn-close" data-bs-dismiss="alert"></button>
</div>
```

Available variants: alert-primary, alert-secondary, alert-success, alert-danger, alert-warning, alert-info, alert-light, alert-dark. Add alert-dismissible + a btn-close button to make it closeable.

3.7 Badges

```
<span class="badge bg-primary">New</span>
<span class="badge bg-danger rounded-pill">99+</span>
```

Small, inline labels for counts or statuses — visually similar to buttons but not clickable/interactive by default. rounded-pill gives fully rounded (pill-shaped) edges.

3.8 Breadcrumbs

```
<nav aria-label="breadcrumb">
  <ol class="breadcrumb">
    <li class="breadcrumb-item"><a href="#">Home</a></li>
    <li class="breadcrumb-item"><a href="#">Library</a></li>
    <li class="breadcrumb-item active" aria-current="page">Data</li>
  </ol>
</nav>
```

Shows the user's current location in a site hierarchy (e.g., Home / Library / Data), similar in markup style to a list — each step is a breadcrumb-item, with the current page marked active.

3.9 Tables

```
<table class="table table-striped table-hover table-bordered">
  <thead class="table-dark">
    <tr><th>#</th><th>Name</th></tr>
  </thead>
  <tbody>
    <tr><td>1</td><td>Example</td></tr>
  </tbody>
</table>
```

Class	Effect
table-striped	Zebra-striped rows
table-bordered	Borders on all sides of cells
table-hover	Highlights a row on mouse hover
table-dark / table-light	Dark or light color scheme for the whole table
table-responsive	Wrap the table in a scrollable container on small screens

3.10 Forms

```
<form>
  <div class="mb-3">
    <label for="email" class="form-label">Email</label>
    <input type="email" class="form-control" id="email">
  </div>
  <div class="form-check">
    <input class="form-check-input" type="checkbox" id="check1">
    <label class="form-check-label" for="check1">Check me out</label>
  </div>
  <button type="submit" class="btn btn-primary">Submit</button>
</form>
```

Class	Purpose
form-control	Styles text inputs, textareas, and selects
form-label	Styles a <label>
form-select	Styles a <select> dropdown
form-check / form-check-input / form-check-label	Checkbox & radio layout
is-valid / is-invalid	Validation state styling (green/red)

4. Utility Classes

4.1 Spacing (Margin & Padding)

Spacing utilities follow the pattern: {property}{sides}-{size}

Segment	Options	Meaning
property	m (margin), p (padding)	Which box-model property to set
sides	t (top), b (bottom), s (start/left), e (end/right), x (left+right), y (top+bottom), blank (all 4 sides)	Which side(s) to apply to
size	0, 1, 2, 3, 4, 5, auto	0 = 0, 1 = 0.25rem, 2 = 0.5rem, 3 = 1rem, 4 = 1.5rem, 5 = 3rem
<pre><div class="mt-3 mb-4 px-2">...</div> <!-- margin-top, margin-bottom, padding-left+right --> <div class="m-0">...</div> <!-- margin: 0 on all sides --> <div class="mx-auto">...</div> <!-- horizontally center a block element --></pre>		

Correction from your notes: the scale is 0 through 5 (not 0–10), and the side letters are t (top), b (bottom), s (start), e (end), x (horizontal), y (vertical) — 's' and 'e' replaced the old 'l'/'r' (left/right) in Bootstrap 4→5 to better support right-to-left languages.

4.2 Typography

<pre><h1 class="display-1">Display 1</h1> <!-- display-1 through display-6, largest to smallest --> <p class="lead">A lead paragraph, slightly larger and lighter.</p> <blockquote class="blockquote"> <p>A well-known quote.</p> <footer class="blockquote-footer">Someone famous</footer> </blockquote></pre>	
Class	Effect
display-1 ... display-6	Extra-large, stylized headings for hero sections
lead	Slightly larger, lighter-weight intro paragraph text
blockquote / blockquote-footer	Styled quotation block with an optional attribution line
text-start / text-center / text-end	Text alignment
text-uppercase / text-lowercase / text-capitalize	Text casing
fw-bold / fw-bolder / fw-normal / fw-light	Font weight
fst-italic	Italic style
text-truncate	Truncates overflowing text with an ellipsis
text-break	Forces long words to wrap instead of overflowing

4.3 Images

```



```

Class	Effect
img-fluid	Makes the image responsive (max-width: 100%, height: auto)
rounded	Slightly rounded corners
rounded-circle	Fully circular (great for avatars)
img-thumbnail	Adds a bordered, padded frame around the image

4.4 Dark Mode

Bootstrap 5.3+ has built-in dark mode support via a data attribute — no separate dark-mode stylesheet needed.

```
<html data-bs-theme="dark">
  <!-- entire page uses dark color mode -->

<div data-bs-theme="dark">
  <!-- just this section uses dark mode -->
</div>
```

5. Common Bootstrap Interview Questions

Note: your ChatGPT share link for interview questions couldn't be retrieved (it returns no extractable content when fetched), so this section is a standard, commonly-asked set compiled independently — not pulled from that specific link.

Q1. What is Bootstrap and why use it?

A free, open-source front-end framework providing pre-built CSS/JS components and a responsive grid system, so developers don't have to build common UI patterns and responsive layouts from scratch.

Q2. What is the difference between container, container-fluid, and container-{breakpoint}?

container has a fixed, responsive max-width per breakpoint; container-fluid is always 100% width; container-{breakpoint} is fluid until that breakpoint, then becomes fixed-width.

Q3. Explain Bootstrap's grid system.

A 12-column, flexbox-based grid built from container > row > col elements. Columns can be sized (col-6), auto-sized (col), or made responsive by combining breakpoint infixes (col-md-4).

Q4. What does 'mobile-first' mean in Bootstrap?

Base CSS rules target the smallest screens by default; min-width media queries progressively enhance the layout for larger screens, rather than starting from desktop and scaling down.

Q5. What are Bootstrap breakpoints?

sm (≥576px), md (≥768px), lg (≥992px), xl (≥1200px), xxl (≥1400px) — used as infixes in grid, spacing, and display utility classes to apply styles at specific viewport widths.

Q6. How do you make a navbar responsive?

Use navbar-expand-{breakpoint} — the navbar collapses into a toggleable hamburger menu below that breakpoint and expands into a horizontal menu at and above it.

Q7. What's the difference between Bootstrap 4 and Bootstrap 5?

Bootstrap 5 drops the jQuery dependency (uses vanilla JS), replaces left/right utilities with start/end (RTL support), adds built-in dark mode (5.3+), and improves the grid with an added xxl breakpoint.

Q8. What are utility classes?

Small, single-purpose classes (e.g., mt-3, text-center, d-flex) that apply one CSS rule each, letting you style elements directly in HTML without writing custom CSS.

Q9. How does the Bootstrap accordion allow multiple open panels?

By omitting the data-bs-parent attribute on each accordion-collapse element — with it set, opening one panel closes the others; without it, multiple panels can stay open independently.

Q10. What is the purpose of data-bs-* attributes?

They're used to wire up Bootstrap's JavaScript-powered components (modals, collapses, carousels, tooltips, etc.) declaratively in HTML, without writing custom JavaScript.

Q11. How do you customize Bootstrap's default styling?

Either override classes with custom CSS loaded after Bootstrap's, or (preferred for larger projects) install via npm/Sass and override Bootstrap's Sass variables before compiling.

Q12. What is the difference between .row and .container?

container sets an outer max-width and centers content; row is a flex wrapper inside a container that holds and lays out the columns, using negative margins to offset column gutters.